Student Exam Performance Analysis

1. Project Overview

The Student Exam Performance Analysis project examines student academic data to understand the factors associated with exam scores and pass rates. The analysis uses Python for data cleaning and analysis and Power BI for dashboard visualization.

2. Dataset Description

- Dataset: student_exam_performance.csv
- Total records: 100,000 students
- Total columns: 44
- Domain: Education

The dataset includes student demographics, study habits, attendance, previous academic performance, exam scores, and pass/fail status.

3. Objectives

- Analyze student exam performance.
- Identify factors related to exam scores.
- Understand the relationship between study hours and marks.
- Examine pass and fail rates.
- Present insights through charts and a dashboard.

4. Data Cleaning

The dataset initially contained 52,161 missing values. Missing numerical values were filled using the median, while missing categorical values were replaced with “Unknown.”

After cleaning:

- Missing values: 0
- Duplicate rows: 0
- Duplicate student IDs: 0

5. Key Findings

- Average exam score: 61.95
- Pass rate: 77.39%

- Students who passed: 77,389
- Students who failed: 22,611

Study hours showed a positive relationship with exam scores. Students studying more hours generally achieved higher average marks.

Previous exam scores and previous GPA were also positively related to current exam performance. Stress level and exam anxiety showed negative relationships with scores.

6. Tools Used

- Python
- Pandas and NumPy
- Matplotlib
- Power BI
- Jupyter Notebook

7. Conclusion

The analysis indicates that study habits and previous academic performance are associated with students' exam results. The project provides useful insights into student performance and demonstrates how data analysis and visualization can support educational decision-making.